

Experiment–1

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Slot: B

Title

Setup and Verification of Anaconda Navigator, Python, TensorFlow, Keras, and Gymnasium for Reinforcement Learning

AIM

To set up the required Python-based software and libraries for Reinforcement Learning and confirm their proper functioning by testing TensorFlow, Keras, Gymnasium, and other supporting packages.

PROCEDURE

1. Download and install the **Anaconda Distribution** on the system.
2. Launch **Anaconda Navigator** or **Anaconda Prompt** after completing the installation.
3. Create an independent Conda environment specifically for Reinforcement Learning applications.
4. Activate the newly created environment through Anaconda Prompt.
5. Check the installed **Python version** to ensure that Python is configured correctly.
6. Install the **TensorFlow** framework using the appropriate pip command.
7. Install **Keras** and test whether it can be imported without errors.
8. Install **Gymnasium**, which contains ready-to-use environments for Reinforcement Learning.
9. Add the supporting packages **NumPy**, **Pandas**, and **Matplotlib** required for numerical processing, data handling, and visualization.
10. Launch **Jupyter Notebook** or use **Google Colab** as the Python programming environment.
11. Import TensorFlow, Keras, Gymnasium, NumPy, Pandas, and Matplotlib.
12. Display the versions of the installed libraries to confirm that they are correctly configured.
13. Initialize a basic Reinforcement Learning environment, such as **CartPole**, using Gymnasium.
14. Reset the CartPole environment and check the initial observation returned by the environment.
15. Perform several randomly selected actions and observe the corresponding environment responses.
16. Capture the required execution screenshots and document the successful installation and testing results.

VISUALIZATIONS

Verification Summary Table

S.No.	Software / Library	Main Function	Verification
1	Anaconda	Provides Python distribution and environment management	Anaconda Prompt/Navigator launched correctly
2	Python	Core programming language used for RL implementation	Python version displayed successfully
3	TensorFlow	Supports machine learning and neural-network computations	TensorFlow imported without errors
4	Keras	Provides a high-level interface for deep learning models	Keras import completed successfully
5	Gymnasium	Supplies standard Reinforcement Learning environments	RL environment initialized successfully
6	NumPy	Performs numerical and array-based operations	NumPy imported successfully
7	Pandas	Used for organizing and processing datasets	Pandas imported successfully
8	Matplotlib	Used to create graphs and visual representations	Plot generated successfully
9	CartPole	Provides a basic environment for testing RL algorithms	Environment reset and actions executed
10	Jupyter Notebook / Google Colab	Platform for writing and running Python programs	Python program executed successfully

RESULT

The required software environment for **Reinforcement Learning** was configured successfully. Python, TensorFlow, Keras, Gymnasium, NumPy, Pandas, and Matplotlib were verified through successful imports and version checks. The **CartPole environment** was also initialized, reset, and tested with sample actions, confirming that the Reinforcement Learning environment is functioning correctly.